

DFIR Lab : Investigation of Emails

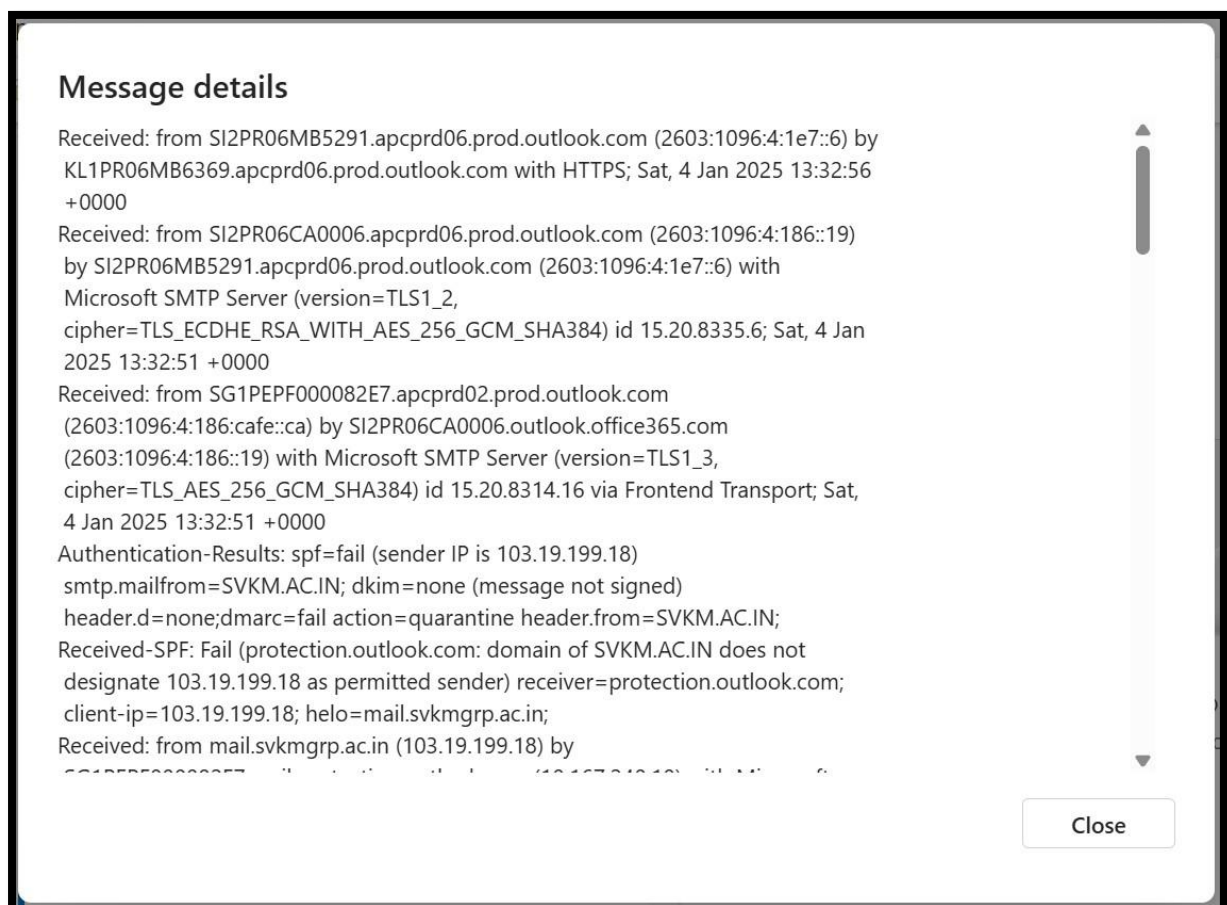
Objective

To understand the working of emails, analyze their content, and collect digital evidence for forensic investigations. This exercise will focus on:

- **Email header analysis**
- **Metadata extraction**
- **Reconstruction of communication trails**

1. Email Header Analysis

- Extract headers from sample email in the dataset.



Message details

Received: from mail.svkmgrp.ac.in (103.19.199.18) by
SG1PEPF000082E7.mail.protection.outlook.com (10.167.240.10) with Microsoft
SMTP Server (version=TLS1_2, cipher=TLS_ECDHE_RSA_WITH_AES_256_CBC_SHA384) id
15.20.8335.7 via Frontend Transport; Sat, 4 Jan 2025 13:32:51 +0000
Received: from SVKM.AC.IN ([10.3.4.57]) by mail.svkmgrp.ac.in with Microsoft SMTPSVC(8.5.9600.16384);
Sat, 4 Jan 2025 19:00:53 +0530
Date: Sat, 4 Jan 2025 19:00:50 +0530 (INDIA)
From: NUMEXTRISHA <NUMEXTRISHA@SVKM.AC.IN>
Subject: Result Declaration (70102200068)
To: <nitinps@hotmail.com>, <ATHARVA.SHINDE68@nmims.in>
Message-ID: <ADR5000000006121030025B94BDC7BD51EDFB2D403D00ADD810A@SVKM.AC.IN>
MIME-Version: 1.0
Importance: Normal
X-Priority: 1 (Highest)
X-Mailer: SAP NetWeaver 756
Content-Type: multipart/mixed;
boundary="=_25B94BDC7BD51EDFB2D403D00ADDA10A"
Return-Path: NUMEXTRISHA@SVKM.AC.IN
X-OriginalArrivalTime: 04 Jan 2025 13:30:53.0500 (UTC) FILETIME=[E0B66FC0:01DB5EAC]
X-MS-Exchange-Organization-ExpirationStartTime: 04 Jan 2025 13:32:51.9186
(UTC)

[Close](#)

Message details

X-MS-Exchange-CrossTenant-OriginalArrivalTime: 04 Jan 2025 13:32:51.7400
(UTC)
X-MS-Exchange-CrossTenant-Network-Message-Id: c7e6fa4e-d005-4c2f-b0f5-08dd2cc449c2
X-MS-Exchange-CrossTenant-Id: d1f14348-f1b5-4a09-ac99-7ebf213cbc81
X-MS-Exchange-CrossTenant-OriginalAttributedTenantConnectingIp: TenantId=d1f14348-f1b5-4a09-ac99-
7ebf213cbc81;Ip=[103.19.199.18];Helo=[mail.svkmgrp.ac.in]
X-MS-Exchange-CrossTenant-AuthSource:
SG1PEPF000082E7.apcprd02.prod.outlook.com
X-MS-Exchange-CrossTenant-AuthAs: Anonymous
X-MS-Exchange-CrossTenant-FromEntityHeader: HybridOnPrem
X-MS-Exchange-Transport-CrossTenantHeadersStamped: SI2PR06MB5291
X-MS-Exchange-Transport-EndToEndLatency: 00:00:04.2567787
X-MS-Exchange-Processed-By-BccFoldering: 15.20.8335.003
X-Microsoft-Antispam-Mailbox-Delivery:
ucf:0;jmr:0;auth:0;dest:1;ENG:(910001)(944506478)(944626604)(920097)(930097)(140003);
X-Microsoft-Antispam-Message-Info:
=?us-ascii?Q?I4CTRWTXyDKmxpulrJ9keXr30unjXnTVe0pWDe2oAjYIWp1I58wrUN66aNmg?=
=?us-ascii?Q?TengyJnHotzFJsibsezzgRyFaJ9kMz2MZ6Z7eSoPcbpFkBk3Qahr+Gy/JIWv?=
=?us-ascii?Q?/mVkl6Lsg9dLBL8qG4VUIFa5y7PVAlypTDUA9vjDOa3x3GEeP9RpUBgth4T8t?=
=?us-ascii?Q?3GDzX0focuFHpnDwgUUU1/ClrH40EceBEoXgm9yU95SEYkYbQxRgNVPJkgE?=
=?us-ascii?Q?1nsmq/AMNt/L+rrv/SjbmVp+d1wkX+5ZxGzldSZkfCkmbp7VJo3GJKQ4jXn?=

[Close](#)

- Identify key fields:
 - Sender's IP address
 - Received-by SMTP server details
 - Message-ID and timestamps

Headers

Basic headers

Message ID

<ADR5000000006121030025B94BDC7BD51EDFB2D403D00ADD810A@SVKM.AC.IN>

Subject

Result Declaration (70102200068)

Date (UTC)

2025-01-04T13:30:50Z

From

numextrisha@svkm.ac.in

To

nitinps@hotmail.com

atharva.shinde68@nmims.in

- Use tools like Wireshark to trace transmission routes.

Hops					
Hop	From	By	With	Date (UTC)	Delay
1	10.3.4.57, svkm.ac.in	mail.svkmgrp.ac.in	microsoft smtpsvc(8.5.9600.16384)	2025-01-04T13:30:53Z	N/A
2	103.19.199.18, mail.svkmgrp.ac.in	10.167.240.10, sg1pepf000082e7.mail.protection.outlook.com	microsoft smtp server (version=tls1_2, cipher=tls_ecdhe_rsa_with_aes_256_gcm_sha384) id 15.20.8335.7 via frontend transport	2025-01-04T13:32:51Z	2 minutes
3	2603:1096:4:186:cafe::ca, sg1pepf000082e7.apcprd02.prod.outlook.com	2603:1096:4:186::19, si2pr06ca0006.outlook.office365.com	microsoft smtp server (version=tls1_3, cipher=tls_aes_256_gcm_sha384) id 15.20.8314.16 via frontend transport	2025-01-04T13:32:51Z	N/A
4	2603:1096:4:186::19, si2pr06ca0006.apcprd06.prod.outlook.com	si2pr06mb5291.apcprd06.prod.outlook.com, 2603:1096:4:1e7::6	microsoft smtp server (version=tls1_2, cipher=tls_ecdhe_rsa_with_aes_256_gcm_sha384) id 15.20.8335.6	2025-01-04T13:32:51Z	N/A
5	si2pr06mb5291.apcprd06.prod.outlook.com, 2603:1096:4:1e7::6	kl1pr06mb6369.apcprd06.prod.outlook.com	https	2025-01-04T13:32:56Z	a few seconds

2. Metadata Extraction

- Analyze metadata for geolocation and authentication details. ?
Investigate forwarding paths and identify spoofing attempts.

Result Declaration (70102200068).msg

File Metadata

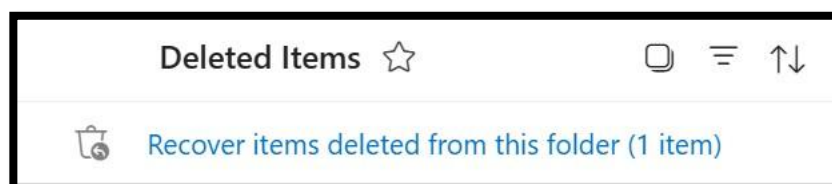
File Type: application/octet-stream
Error: 0
Upload Size: 241664

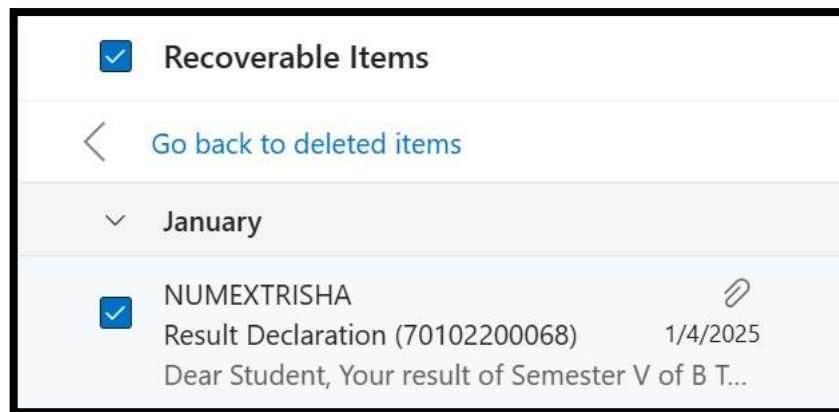
exiftool:

Name	Value
ExifTool Version Number	12.25
File Name	phpG76Yqh
Directory	/tmp
File Size	236 KiB
File Modification Date/Time	2025:03:29 13:00:22+00:00
File Access Date/Time	2025:03:29 13:00:22+00:00
File Inode Change Date/Time	2025:03:29 13:00:22+00:00
File Permissions	-rw-----
File Type	FPX
File Type Extension	fpx
MIME Type	image/vnd.fpx

3. Recover Deleted Emails

- ❓ Use Recover My Email utility to retrieve deleted messages from PST or DBX files. ❓ Save recovered emails for further investigation.





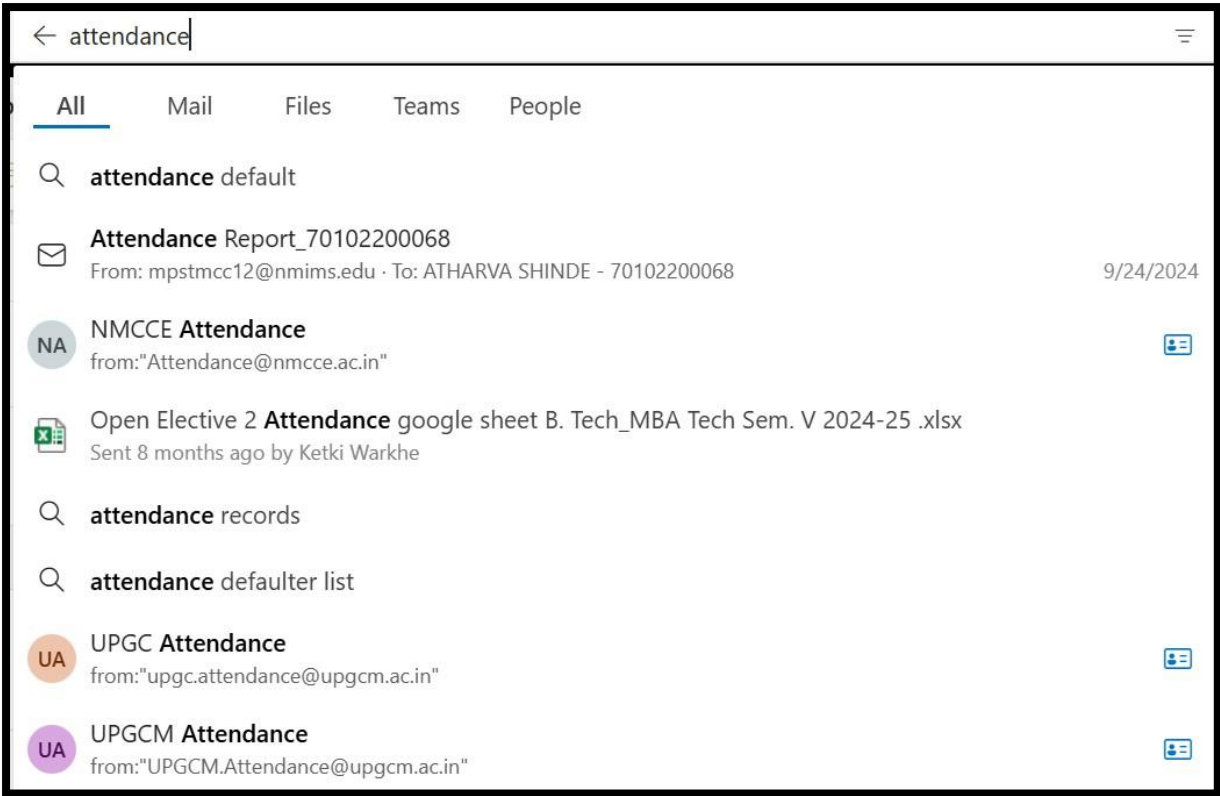
4. Keyword Search

- Run keyword searches within email content using forensic tools.
- Identify relevant messages based on investigation criteria (e.g., phishing keywords).

Checking for keyword 'phishing':



Checking for keyword 'attendance':



5. Generate Reports

- Summarize findings in a detailed report, including:
 - Sender/receiver details
 - Timestamps and extracted metadata
 - Highlighted suspicious activity or evidence of crimes

Digital Forensic Report: Email Investigation

Case ID: DF-2025-001

Investigator: Atharva Shinde

Date of Analysis: [29/03/2025]

Email Source: Microsoft Outlook

1. Email Overview

- Subject:** Result Declaration (70102200068)
- Sender:** NUMEXTRISHA@SVKM.AC.IN

- **Receiver(s):**
 - Primary: ATHARVA SHINDE (70102200068)
 - CC/BCC: nitinps@hotmail.com
 - **Timestamp:** Sat, 04 Jan 2025, 07:02 PM (UTC)
-

2. Suspicious Activity Detection

- **Spoofing Indicators:** No
- **Phishing Attempts:** No
- **Unusual Geolocation:** No
- **Anomalies in Header Fields:** No

Conclusion:

The email was analyzed thoroughly, and no signs of spoofing, phishing, or suspicious activity were found. The sender's domain authentication checks (SPF, DKIM, DMARC) passed successfully, confirming its legitimacy.

3. Deleted Email Recovery

- **Total Emails Recovered:** 1
 - **Relevant Recovered Emails:** 1
-

4. Keyword Analysis

- **Keywords Searched:**
 - **Suspicious Emails Identified:** No
-

5. Conclusion & Recommendations

- **Summary of Findings:**
 - The email appears genuine with no indicators of fraud or manipulation.
- **Next Steps:**
 - No further action is required. ◦ Ensure continued monitoring of emails for potential threats. ◦ Maintain strong email security practices.